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A Sensitivity Study on Aeroelastic Instabilities of Slender Wings with a Large Propeller

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Next-generation aircraft designs often incorporate multiple large propellers attached along the wingspan. These highly flexible dynamic systems can exhibit uncommon aeroelastic instabilities, which should be carefully investigated to ensure safe operation. The interaction between the propeller and the wing is of particular importance. It is known that whirl flutter is stabilized by wing motion and wing aerodynamics. This paper investigates the effect of a propeller onto wing flutter as a function of span position and mounting stiffness between the propeller and wing. The analysis of a comparison between a tractor and pusher configuration has shown that the coupled system is more stable than the standalone wing for propeller positions near the wing tip for both configurations. The wing flutter mechanism is mostly affected by the mass of the propeller and the resulting change in eigenfrequencies of the wing. For very weak mounting stiffnesses, whirl flutter occurs, which was shown to be stabilized compared to a standalone propeller due to wing motion. On the other hand, the pusher configuration is, as to be expected, the more critical configuration due to the attached mass behind the elastic axis.

I. Nomenclature

Latin Symbols

e	=	distance propeller hub - pivoting point $[m]$
EI	=	bending flexibility $[Nm^2]$
F	=	force $[N]$
GJ	=	torsional flexibility $[Nm^2]$
J	=	advance ratio [-]
k	=	reduced frequency [-]
K_θ	=	rot. stiffness around pitch $[Nm]$
K_ψ	=	rot. Stiffness around yaw $[Nm]$
M	=	Moment $[Nm]$
Ma	=	Mach Number [-]
ρ	=	air-density $[kg \cdot m^{-3}]$
V	=	air-velocity $[m \cdot s^{-1}]$
θ_P	=	generalized coordinate for the propeller in pitch $[rad]$
ψ_P	=	generalized coordinate for the propeller in yaw $[rad]$
θ_W	=	generalized coordinate for the wing in pitch $[rad]$
h	=	generalized coordinate for the wing in heave $[m]$

Subscripts

W	=	wing
P	=	propeller
PW/WP	=	propeller-wing coupling

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II. Introduction

Electrification has brought many advantages to aircraft designers in terms of propulsion integration. Next-generation configurations can take advantage of electric propulsion in the form of aerodynamic efficiency, simplicity, and low noise. The flexibility of placing the propulsion in beneficial locations and the relatively small weight of electric propulsion led to the concept of distributed electric propulsion systems (DEPs). Because of the propellers' high efficiency at low speed, it is not surprising that the choice of the propulsion system turns to large propellers distributed along the wingspan. Even though many concepts feature DEPs in classical tractor configuration, pusher configurations are not excluded in principle.

For the sake of aerodynamic efficiency, lifting surfaces of these novel configurations are often designed with high aspect ratios. These mass-optimized structures are highly flexible by nature. This flexibility, in combination with large rotating propellers, makes dynamic behavior a critical factor, especially in aeroelasticity for aircraft design. By looking at the individual parts of this configuration, the critical aeroelastic phenomena are expected to be classical wing flutter and propeller whirl flutter. Both dynamic aeroelastic instabilities must be avoided since they would cause fatal accidents. It is therefore required by certification specifications to show compliance with a flutter-free aircraft within the operational envelope.

Aeroelastic instabilities for the individual parts (wing and propeller) have long been investigated for classical wing flutter and propeller whirl flutter. Classical wing flutter is usually modeled using a simplified structural model of a flexible wing and appropriate unsteady aerodynamics theories, such as the Doublet-Lattice-Method (DLM). The classical whirl-flutter theory was developed for a propeller flexible attached in yaw and pitch at its pylon, assuming the propeller blades as rigid. Propeller aerodynamic models were developed depending on blade geometry for windmilling propellers in yaw and pitch [1, 2].

The interactions between the wing and propeller are modeled and are of particular interest. For a coupled wing-propeller model, several studies also have been conducted, beginning with an analytical and experimental approach by Bennett and Bland in 1994 [3]. Due to limitations in numerical calculations, their approach was simplified to a reduced method employing the Ryleigh-Ritz approach. Moreover, it is well known that even discrete masses influence the aeroelastic behavior of aircraft wings [4]. This influence is intensified by the gyroscopic effects of rotating masses [5, 6], the thrust produced by the propellers [7], and the aerodynamic effects of propellers on the flow around the wing [8]. Further, it is known that wing motion has a stabilizing effect in terms of classical whirl flutter. This is often indicated by typical stability boundary curves for a propeller, as qualitatively shown in Fig. 1. The mounting stiffnesses describe the propeller attachment flexibility about a pivoting point in yaw and pitch. The theoretical case of a standalone propeller (curve A) is shown to be the most critical generating the largest unstable boundary. Wing flexibility (curves B and C) stabilizes the whirl flutter, except in a region with unsymmetrical mounting stiffnesses (shadowed area). Nevertheless, these correlations are only available for the whirl flutter stability and a tractor configuration.

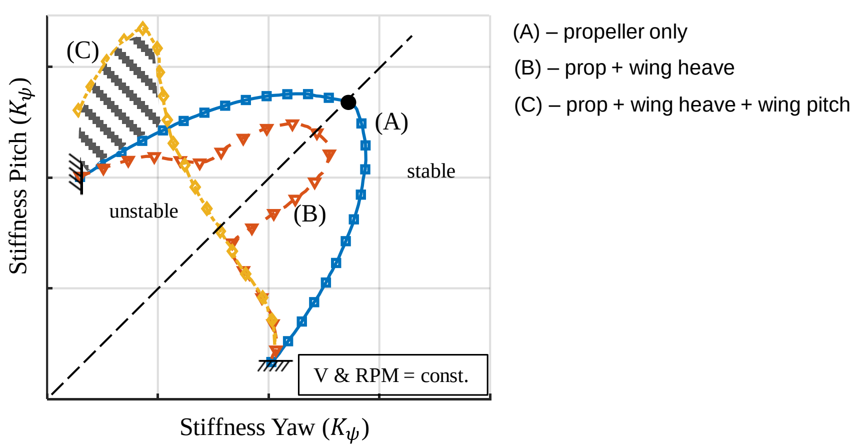


Fig. 1 Whirl flutter boundary and dependencies on wing motion given in the literature [1, 9]

Several questions remain unanswered. For example, it is not clear how flexible propeller blades influence classical wing flutter. This is an essential aspect since a weak propeller mounting onto a wing is usually nonexistent. Of course, in the case of a structural failure, whirl flutter may be present and thus needs to be investigated by means of a failure case analysis with reduced mounting stiffness. However, the normal flutter phenomenon is believed to be wing flutter.

The second question refers to pusher configuration, where the propellers are attached aft of the elastic axis of the wing, which will foster aeroelastic instabilities. Thus detailed knowledge of complex wing-propeller dynamic systems is not available, although several studies have been performed recently [10, 11]. To close this gap would be beneficial for aircraft design engineers to account for dynamic aeroelastic instabilities during preliminary aircraft design.

This paper addresses how an attached propeller influences wing flutter for various span positions and mounting stiffnesses. In addition, a comparison between tractor and pusher configurations will be performed. The studies are intended to provide an understanding of aeroelastic instabilities and their origins that can occur in complex dynamic systems.

III. Methodology

A. Coupled Wing-Propeller Model – Frequency Domain:

The system under consideration is a coupled wing-propeller model. The coupled model's top- and cross-sectional views are depicted in Fig. 2. The wing's structural parameters are based on a Bernoulli beam allowing for out-of-plane bending and torsion displacements. Thus, each wing element node (n) has three degrees of freedom (DoFs), $\boldsymbol{q}_W^n = \{h, \varphi, \theta_W\}$. The propeller will be flexibly attached to the wing with two rotational springs (pitch and yaw), leading to two additional DoFs for the propeller, $\boldsymbol{q}_P = \{\theta_P, \psi_P\}$. The attachment (pivoting point – PP) is located at a distance lgt from the elastic axis of the wing, and the propeller hub is located at a distance e fore or behind the PP (see Fig. 2). The shaft and propeller blades are assumed to be rigid.

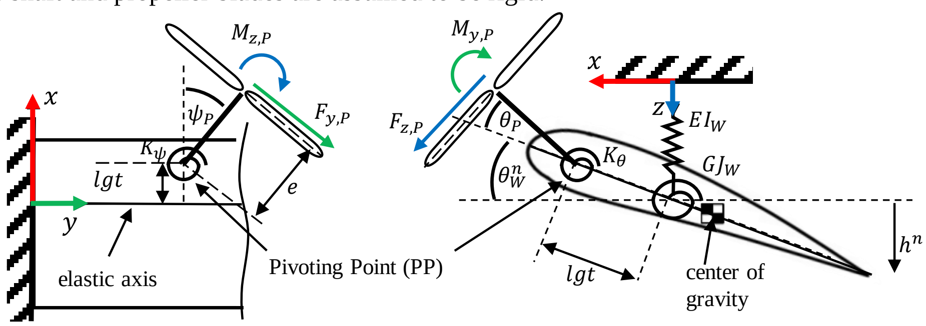


Fig. 2 Schematic sketch of the dynamic wing-propeller model (according to [3])

The coupled wing-propeller model can then be described in terms of the well-known second-order differential equations of motion:

$$\begin{bmatrix} \boldsymbol{M}_W + \boldsymbol{M}'_{WP} & \boldsymbol{M}''_{WP} \\ \boldsymbol{M}''_{PW} & \boldsymbol{M}_P \end{bmatrix} \begin{Bmatrix} \ddot{\boldsymbol{q}}_{\boldsymbol{w}} \\ \ddot{\boldsymbol{q}}_{\boldsymbol{P}} \end{Bmatrix} + \begin{bmatrix} \boldsymbol{C}_W & \boldsymbol{C}_{WP} \\ \boldsymbol{C}_{PW} & \boldsymbol{C}_P \end{bmatrix} \begin{Bmatrix} \dot{\boldsymbol{q}}_{\boldsymbol{w}} \\ \dot{\boldsymbol{q}}_{\boldsymbol{P}} \end{Bmatrix} + \begin{bmatrix} \boldsymbol{K}_W & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{K}_P \end{bmatrix} \begin{Bmatrix} \boldsymbol{q}_{\boldsymbol{w}} \\ \boldsymbol{q}_{\boldsymbol{P}} \end{Bmatrix} = \begin{Bmatrix} \boldsymbol{Q}_{q_W} \\ \boldsymbol{Q}_{q_P} \end{Bmatrix} \tag{1}$$

Herein, the generalized coordinates $\boldsymbol{q}$ are a combination of the wing degrees of freedom ($\boldsymbol{q}_W$) and the propellers degrees of freedom ($\boldsymbol{q}_P$). The structural mass ($\boldsymbol{M}$), damping ($\boldsymbol{C}$), and stiffness ($\boldsymbol{K}$) matrices are formulated by coupling terms due to the mounting of the propeller. The detailed definition of these matrices is in [12]. It is worth noting that only the inertia and damping matrix contains coupled terms between the wing and propeller. The former is because an additional mass is attached to the flexible wing. Damping couplings occur because the rotating propeller leads to precession, and this gyroscopic motion couples the propeller pitch DoF with the wing's pitch DoF as described in [12]. No coupling appears in the stiffness.

The generalized forces $\boldsymbol{Q}$ in Eq. (1) contain the aerodynamic forces acting on the structural nodes. No other external excitations are considered. Then, the aerodynamic forces consist of the wing and propeller contributions and can be formulated as follows:

$$\begin{Bmatrix} \boldsymbol{Q}_{q_W} \\ \boldsymbol{Q}_{q_P} \end{Bmatrix} = \begin{Bmatrix} \boldsymbol{Q}_{WP} \\ \boldsymbol{Q}_P \end{Bmatrix} + \frac{1}{2}\rho V^2 \begin{bmatrix} \Re\big(\boldsymbol{AIC}(k, Ma)\big) & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix} \begin{Bmatrix} \boldsymbol{q}_{\boldsymbol{w}} \\ \boldsymbol{q}_{\boldsymbol{P}} \end{Bmatrix} + \frac{1}{2}\rho V \frac{b}{k} \begin{bmatrix} \Im\big(\boldsymbol{AIC}(k, Ma)\big) & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix} \begin{Bmatrix} \dot{\boldsymbol{q}}_{\boldsymbol{w}} \\ \dot{\boldsymbol{q}}_{\boldsymbol{P}} \end{Bmatrix} \tag{2}$$

The last two summands describe the unsteady wing aerodynamic forces through the aerodynamic influence coefficients (AIC). Those are dependent on the reduced frequency k and Mach number Ma. The AIC matrices are calculated using the DLM with the open-access tool DLMPro[5] developed by the authors [13]. The imaginary and real parts are then used to calculate the aerodynamic damping and stiffness matrices, respectively.

The first summand in Eq. (2) describes the generalized aerodynamic forces generated by the propeller. In contrast to the wing aerodynamics, the propeller resulting forces influence the wing's DoFs as well as the propeller's DoFs. $\boldsymbol{Q}_P$ describes the moments acting on the PP, according to the coordinated system defined in Fig. 2:

$$\boldsymbol{Q}_P = \begin{Bmatrix} -e \cdot F_{z,P} + M_{y,P} \\ e \cdot F_{y,P} + M_{z,P} \end{Bmatrix} \tag{3}$$

The propeller forces and moments are calculated using the propeller stability derivates from [2]. They are valid for windmilling conditions. In general, they describe the change of the forces and moments depending on the effective deflection angle $\tilde{\theta}$ and $\tilde{\psi}$. Due to the coupling with the wing, those effective angles are defined for the coupled model as:

$$\begin{aligned} \tilde{\theta} &= \theta_P - \frac{e_P}{V}\dot{\theta}_P + \theta_W^n - \frac{e_W}{V}\dot{\theta}_W^n + \frac{1}{v}\dot{h}^n \\ \tilde{\psi} &= \tilde{\psi}_P - \frac{e}{V}\dot{\psi}_P \end{aligned} \tag{4}$$

Due to these coupled effective angles, the vector $\boldsymbol{Q}_{WP}$ describes the generalized forces of the propeller aerodynamic acting on the wing DoF(s). It can be shown that the propeller forces and moments depend on both, the n-th wing node DoFs (where the propeller is attached) and propeller DoFs and their derivatives. Thus, the propeller forces and moments couple the wing and propeller DoFs as well as indicated in Eq. (5). For a more detailed overview, the reader is referred to [12].

$$\boldsymbol{Q}_{WP} = f({\boldsymbol{q}_W^n}^T, {\dot{\boldsymbol{q}}_W^n}^T, {\ddot{\boldsymbol{q}}_W^n}^T, {\boldsymbol{q}_P}^T, {\dot{\boldsymbol{q}}_P}^T, {\ddot{\boldsymbol{q}}_P}^T) \tag{5}$$

The coupled aeroelastic wing-propeller model can then be solved using the p-method in combination with the iterative p-k-method. The routines are performed in an in-house MATLAB tool called SDBox. Stability analyses are performed for a freestream velocity range of interest. For each velocity, the eigenvalue problem will be solved iteratively since the eigenvalues also depend on the wing aerodynamics. The propeller aerodynamic model does not depend on the reduced frequency and will be superimposed by means of a classical p-method. Aerodynamic interaction between the propeller and wing is neglected. The propeller derivatives are calculated according to [2, 3], and the AICs are calculated with DLMpro. The aeroelastic analysis will result in typical flutter curves, showing the mode - frequencies and damping values versus airstream velocity.

B. Description of the study performed

This paper focuses on the comparison between a tractor and pusher configuration. An overview of the performed study is depicted in Fig. 3. The coupled model will be formed using a baseline wing model and a propeller model. The propeller will be flexibly attached to the wing. The models are unchanged, but the spanwise position of the propeller and the mounting stiffness between the wing and propeller will vary. These variations are done for both configurations, tractor and pusher. An aeroelastic analysis is performed for each system, considering inertia effects and wing- and propeller aerodynamics. Aerodynamic interactions between the wing and propeller are neglected. The analysis results will give critical aeroelastic velocities and corresponding critical flutter mode shapes. They are used to compare the different systems and both configurations to each other with respect to dynamic stability.

[5] https://github.com/nilsBoeAc/DLMpro

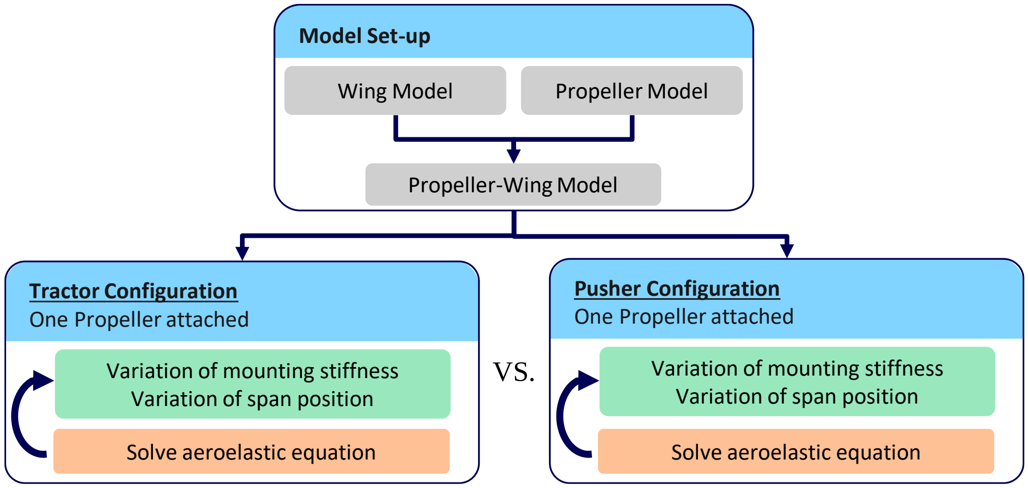


Fig. 3 Overview of the conducted research

IV.Model Description

C. Baseline Wing Model

The baseline wing used in this paper is the same as used in the studies performed by the authors in Ref. [6]. This rectangular wing with an aspect ratio of 9.12 and a half-span of 5.7m has its elastic axis (EA) located at 35% chord from the leading edge. The center of gravity axis (CG) is located 10% chord behind the EA. The total mass of the wing is 102 kg, according to typical values for an aircraft with 2,500 kg maximum take-off mass (MTOM). The bending and elastic stiffnesses of the wing are $29 \cdot 10^5 Nm^2$ and $2.7 \cdot 10^5 Nm^2$, respectively. The corresponding wind-off mode shapes can be found in [6]. A classical flutter analysis was performed using the SDBox for the standalone wing (see Fig. 4 a, b).

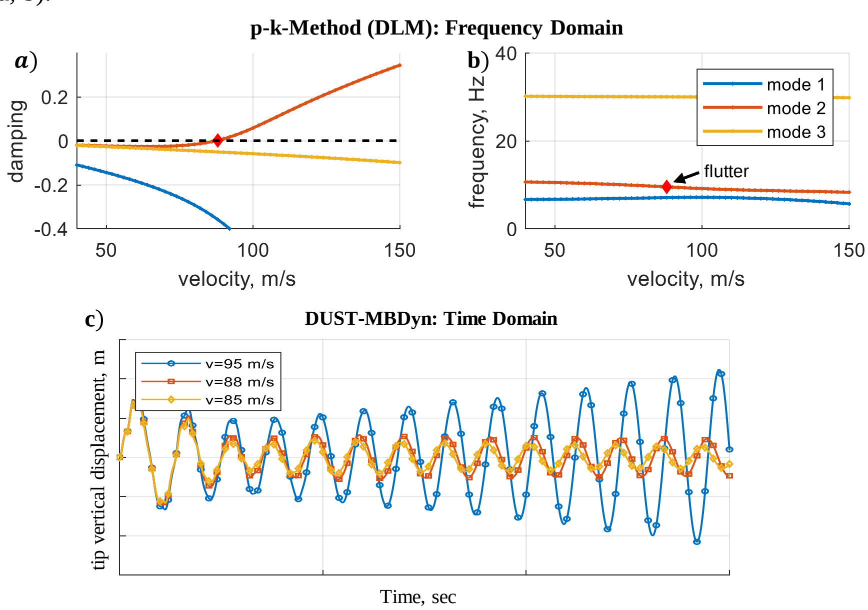


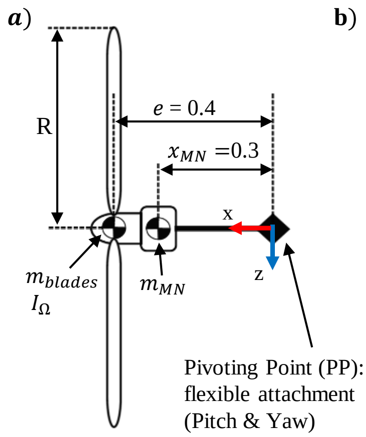
Fig. 4 Flutter curves (a - damping) and (b - frequency) calculated with SDBox using the p-k method in the frequency domain and (c) time-domain results calculated with DUST/MBDyn

Structural damping was neglected. A flutter velocity of 88 m/s is estimated. For validation, time-domain simulations were performed using the mid-fidelity aerodynamic software DUST [14, 15] and the multibody tool

MBDyn [16] (both developed at Politecnico di Milano, coupled with PRECISE[6]). Three different freestream velocities (85 m/s, 88 m/s, 95 m/s) are used, and the dynamic response was simulated using a tight coupling scheme. The wing tip response in the vertical direction is given for all freestream velocities (see Fig. 4c). An unstable behavior is seen for 95 m/s, while for 88 m/s, an undamped behavior is visible. This excellently matches the estimation made with the p-k-method and aerodynamics from DLM.

D. Baseline Propeller Model

The baseline propeller used in this study is a 4-bladed propeller driven by an electric motor. The entire unit string and the blades are assumed to be rigid. A linear twist distribution is assumed. Due to the electrical drive, two mass points are assumed in the propeller, as indicated in Fig. 5. The first mass is dedicated to the rotational part and contains the inertia parameter of the blades. The mass moment of inertia around the spinning axis is estimated, assuming a constant mass distribution of the propeller blades. The advance ratio in windmilling conditions was calculated using DUST. In order to evaluate whirl flutter, the propeller system is attached flexibly to the pivoting point. Two degrees of freedom are modeled, allowing the propeller to rotate in pitch (θ) and yaw (ψ) direction (see [12]). Rotational springs are used throughout this study, and the corresponding stiffness constants (K_θ and K_ψ) are varied, even though they are always equal for pitch and yaw.



b)

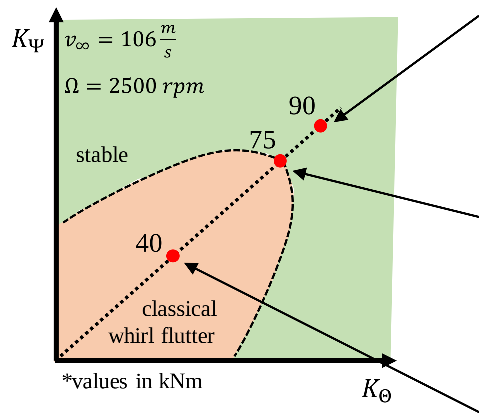
Description	Value
$number\ of\ blades\ [-]$	4
$radius\ (R)\ [m]$	0.9
$blades\ mass\ (m_{blades})\ [kg]$	8.8
$mass\ moment\ of\ inertia$ $(around\ spinnging\ axis\ I_\Omega)\ [kgm^2]$	2.38
$motor\ mass\ (m_{MN})\ [kg]$	25
$e\ [m]$	±0.4 *
$c_{c.g\ (MN)}\ [m]$	±0.3 *
$advance\ ratio\ (J)$ $(windmilling)\ [-]$	1.42

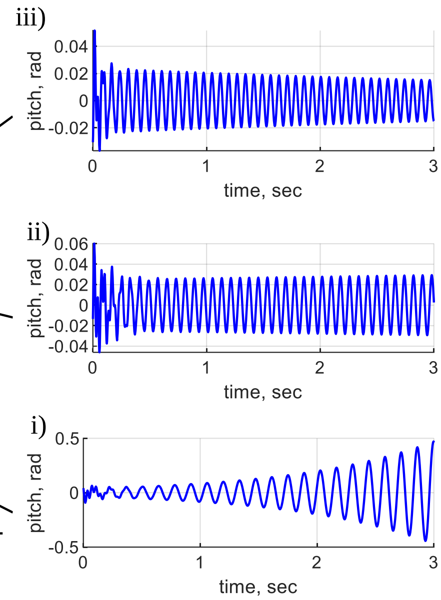
* +: *tractor configuraiton*
−: *pusher configuration*

Fig. 5 Schematic propeller model (a) and propeller data (b)

The standalone propeller was validated using DUST/MBDyn for one freestream velocity. Since the conventional method to estimate whirl flutter is only valid for windmilling conditions, the advance ratio for all simulations is kept constant. The freestream velocity and rotational speed are kept constant as well. For this parameter set, only the mounting stiffnesses are changed. In other words, the required stiffness constant is searched to obtain stability. The boundary indicated in the figure is shown qualitatively. However, the diagonal represents values for equal pitch and yaw stiffness, which is the most critical combination for the classical whirl flutter of a standalone propeller. Using the SDBox, the required stiffness to avoid classical whirl flutter was estimated for the propeller at hand, with a freestream velocity of 106 m/s and a rotational speed of 2,500 rpm to be 65,000 Nm/rad. For validation, three coupled time-domain simulations were performed using different mounting stiffnesses. The response of the various simulations is visualized for the propeller's pitch degree of freedom. It is seen that the case with the lowest stiffness (i) is unstable, whereas the mid-stiffness (ii) and highest stiffness (iii) are undamped and stable, respectively. As seen in Fig. 6, equal dependencies were found. However, DUST/MBDyn is more conservative than SDBox since a higher stiffness is required for neutral stability.

[6] https://precice.org





SDBox $k_\theta = k_\psi$	DUST & MBDyn $k_\theta = k_\psi$	Deviation [%]
65 kNm	75 kNm	15

Fig. 6 Propeller from calculated in DUST-MBDyn and deviation

Since the focus will be on the difference in tractor and pusher configurations, these configurations are investigated for the standalone propeller. As indicated in Fig. 5 (right), the only difference is the location of the additional masses with respect to the pivoting point. Therefore, the propeller force acting on the propeller hub produces a contrary moment compared to the pusher configuration (see equation (3)). In addition, the effective angle of attack changes as well (see equation (4)). Because of this, the aerodynamic response also changes the whirl flutter speed. This comparison can be seen in Fig. 7. The primary axis (solid lines) indicates the whirl flutter velocities, and the secondary axis (dashed lines) indicates the corresponding rotational speed. Keep in mind that the advance ratio is kept constant to maintain windmilling conditions, and thus the rotational speed is changed. The reduced whirl flutter speed for the pusher configuration is clearly visible compared to the tractor configuration, indicating a destabilization for those configurations.

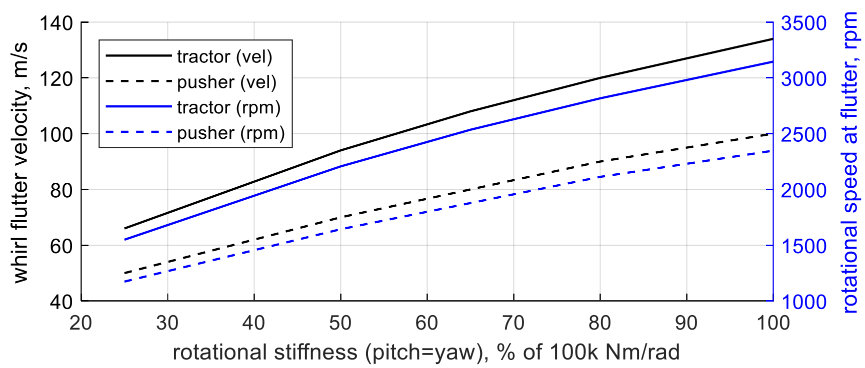


Fig. 7 Whirl flutter velocities (primary axis – black lines) and rotational speed at flutter (secondary axis – blue lines) in comparison between tractor (solid lines) and pusher (dashed lines) configuration for different attachment stiffnesses.

E. Coupled Wing-Propeller Model

The coupled wing-propeller model consists of the baseline wing and one attached baseline propeller. The models described before are used. The placement of the propellers is either in front of the leading edge or behind the trailing edge to yield a tractor (Fig. 8a) or pusher (Fig. 8b) configuration, respectively. The pivoting point of the propeller is attached to the wing's elastic axis with L (tractor) or L' (pusher) from the EA. This length is different for both configurations. However, the gap between the propeller's hub and the wing's geometry is equal for both configurations. Please note that for the first parameter study, only one propeller is attached to the wing. For a rotating propeller and a flexible attachment, the wind-off mode shapes of these coupled systems are essential to characterize the dynamic behavior of the system. Even though they vary with different span positions, mounting stiffness, and the number of propellers, the following figure compares the first 4 wind-off mode shapes. Their primary purpose is to give an idea of what the complex forms of the modes look like. Because of the rotation and the gyroscopic effect, the propeller always performs a whirl motion. Table 1 summarizes labels used for the mode shape description. They can be distinguished between wing and propeller modes. However, the wing motion description does not mean that exclusively this motion is present. This should be clear since the CG has an offset from the EA. The description is meant to describe the primary motion of the wing. Both labels are used if the primary motion cannot be identified clearly. The propeller whirl motion can be either in-phase or out-of-phase to the wings motion. The former occurs when the propeller hub reaches its maximum positive pitch and when the wing reaches its maximum positive pitch as well. It should be noted that positive pitch angles for the wing and propeller are defined in the positive sense of rotation around the y-axis (right-hand rule).

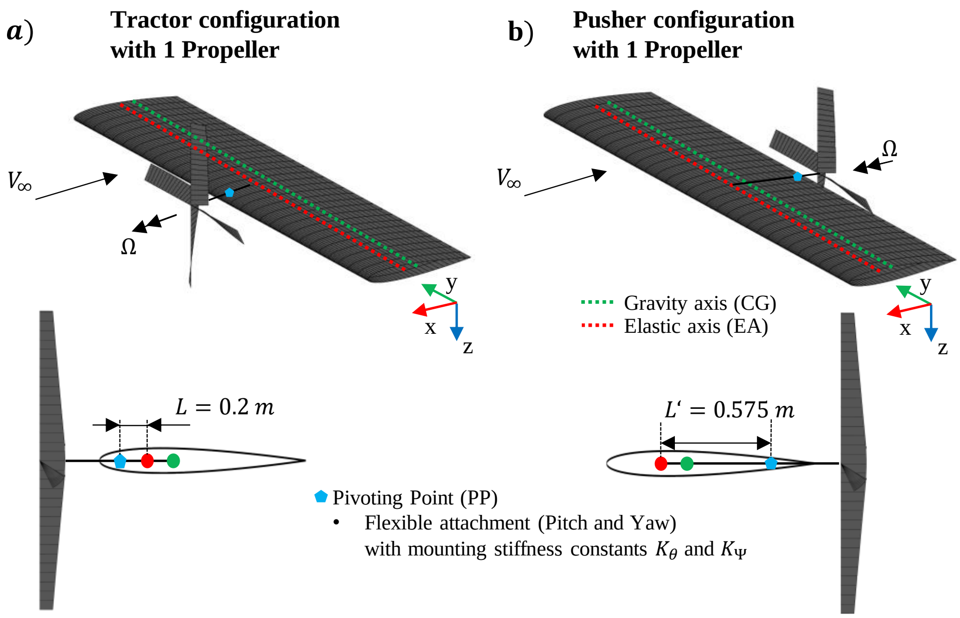


Fig. 8 Tractor (a) and Pusher (b) configuration with one propeller attached at 50% half-span

The wind-off mode shapes for tractor and pusher configurations with one propeller, flexible attached at 50% half-span, are given in Fig. 9. The comparison between the configurations clearly shows a strong coupling between wing bending and torsion for the pusher configuration. This was expected since the propeller is attached behind the elastic axis and therefore amplifies the center of gravity offset.

Table 1 Labels used for mode shape description

Label	Description
n^{th} B	n^{th} Wing Bending (dominant)
n^{th} T	n^{th} Wing Torsion (dominant)
BW (iP)	Backward whirl in-phase with wing motion
BW (oP)	Backward whirl out-of-phase with wing motion
FW	Forward Whirl

For the tractor configuration, a backward whirl motion often appears in conjunction with a wing torsion dominant motion. Aeroelastic analyses are performed for various complex models, including the flexible attachment to the wing. The authors comprehensively validate these models with DUST/MBDyn in [17]. An excellent agreement in flutter velocity and flutter frequency was found between the classical flutter methods in the frequency domain and the mid-fidelity methods in the time domain.

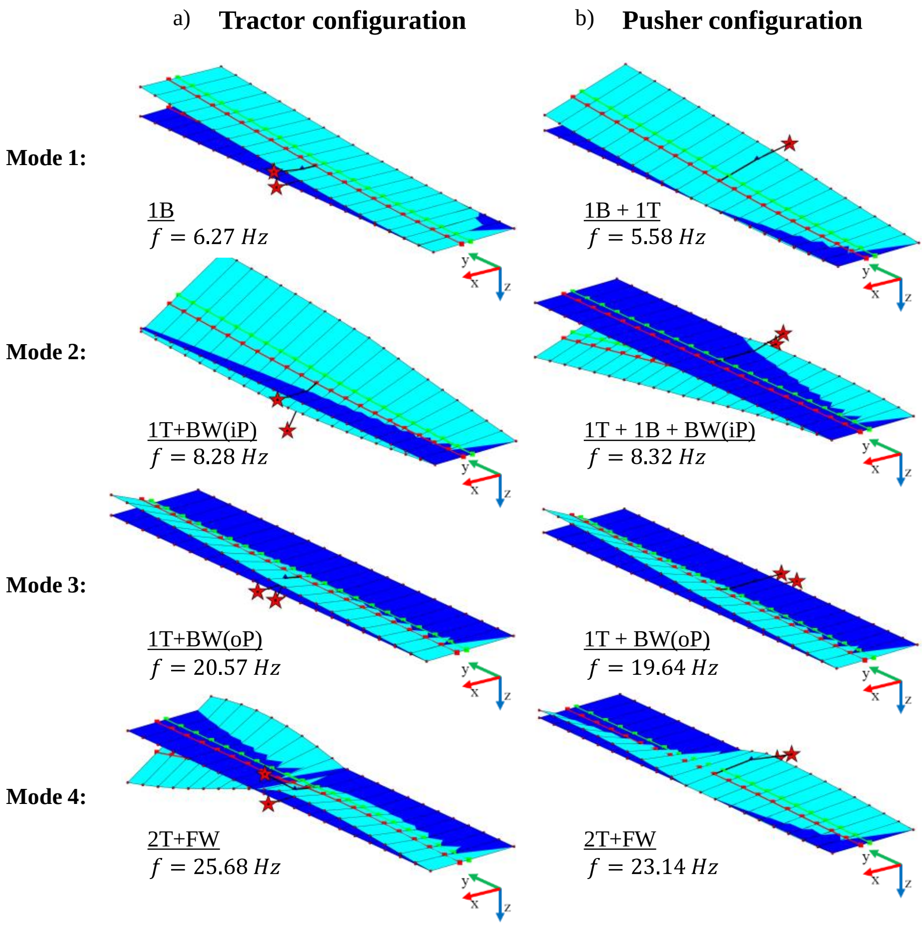


Fig. 9 Wind-off mode shapes for tractor (left) and pusher (right) configuration, including description. Propeller is flexible attached at 50% half-span (red star indicates the propeller hub).

V. Results and Discussion

F. Parameter Study Results: One attached propeller for pusher and tractor configuration

In the performed parameter study, one propeller will be attached to the wing in tractor and pusher configuration. For each configuration, the span position and mounting stiffness will be varied. The latter is always referred to in percent to a strong mounting with a stiffness constant of 100,000 Nm/rad. An aeroelastic analysis is performed for each combination of the variable parameter. The critical aeroelastic velocities are extracted and used to compare the different systems concerning dynamic stability. The flutter mechanism can either be assigned due to wing flutter or classical whirl flutter. Nevertheless, the corresponding flutter mode shape always contains the contribution of wing and whirl motions, independently of the flutter mechanism.

The results for the tractor configuration are summarized in Fig. 10 as a function of span position and mounting stiffness. Mainly, instabilities occur due to the wing flutter, except in the case of the weak mounting of the propeller (Fig. 10 – blue line). In this case, flutter is initiated by propeller whirl motion independent of the spanwise position, which is why the flutter speed is drastically reduced compared to the other mounting stiffnesses. This effect was expected since a weak mounting will foster whirl flutter. With respect to the whirl flutter speed of the standalone propeller (Fig. 10 – dotted line), the whirl flutter speed of the coupled system is always higher. Thus, the flexible wing has a stabilizing effect on whirl flutter, which is in accordance with the literature [9, 18]. With respect to the standalone wing (Fig. 10 – dashed line), the coupled system is destabilized for inboard positions (less than 40% half-spanwise position) of the propeller and stabilized for outboard positions (higher than 40%) of the propeller. For the stronger mountings, the flutter mechanism can be attributed to the wing. However, depending on the span position, the flutter mode changes.

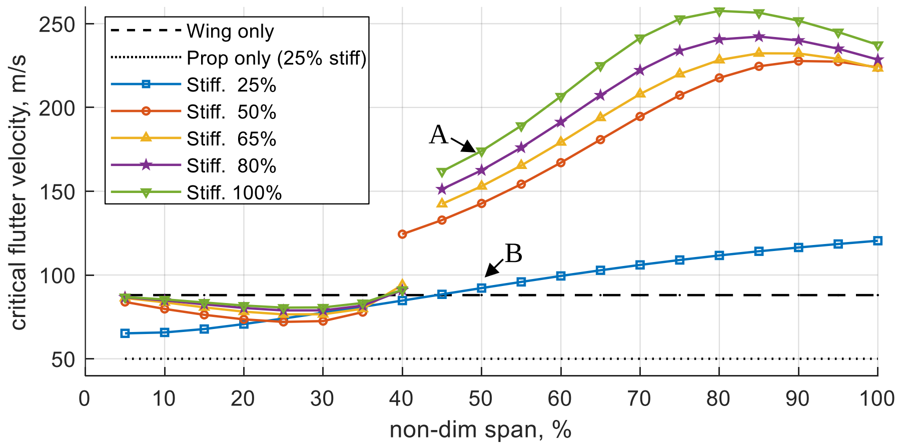


Fig. 10 Critical velocity for a tractor configuration with one attached propeller as a function of span position and mounting stiffness

The change of flutter mode leads to an increase in flutter speed around 40% half-span position. For inboard stations, less than 40% half-span, flutter occurs in mode 2 (1T+BW(iP)), while for half-span positions outer than 40%, the flutter mode is dedicated to mode 3 (1T+BW(oP)). These observations are in accordance with previous studies performed by the authors [12]. The change of flutter mode is due to the fact of slightly changed mode shapes of the system due to the different span positions, similar as explained in [4, 12].

For the outer span positions, the effect of the mounting stiffness on the critical speed becomes significant. The weaker the mounting stiffness, the less the critical flutter velocity. Corresponding flutter curves and flutter mode shapes for two different mounting stiffnesses (Fig. 10 – Point A and B) for an attached propeller at 50% half-span are shown in Fig. 11. For the case of the very weak mounting (B), the flutter mode shape is dominated by the backward whirl (BW), which is in phase with the wing's pitch. This flutter mechanism is referred to whirl flutter, which is expected due to the weak mounting stiffness.

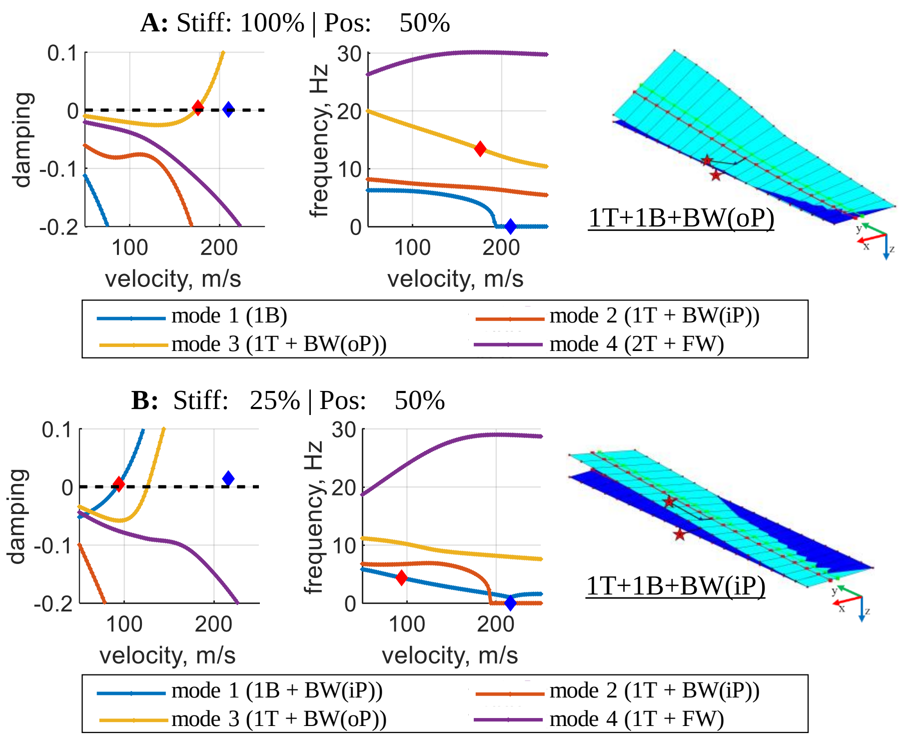


Fig. 11 Flutter curves and flutter mode shape for tractor configuration with the propeller at 50% half-span for two different mounting stiffnesses. Red diamond shows flutter. Blue diamond shows divergence.

The flutter mode shape for the highest mounting stiffness (A) is representative for all other flutter mode shapes for positions higher than 40% half-span and mounting stiffnesses higher 50%. The smaller wind-off eigenfrequency spacing between mode 2 and mode 3 (see Fig. 12) may explain the lower critical flutter speed for reduced stiffnesses, which is more prone to coupling. Considering the propeller's span position, the system is stabilized with increasing span position (expect a slight decrease for positions near the wing tip).

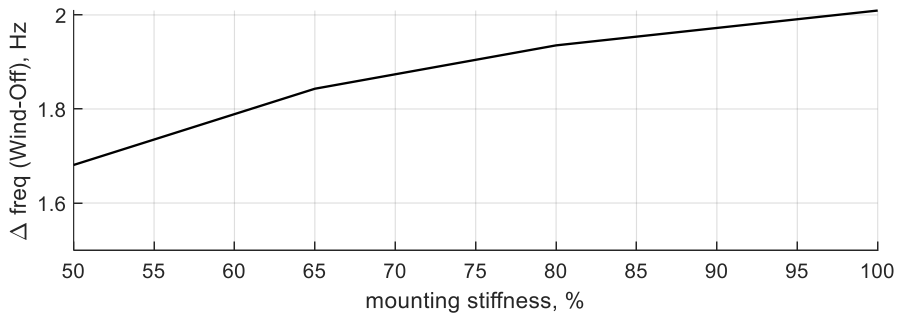


Fig. 12 Difference (abs) in wind-off eigenfrequencies between mode 2 and 3 for 50% half-span position

A similar study was performed for the pusher configuration, which is much more critical in terms of aeroelastic stability since the propeller is attached behind the elastic axis. The offset between the center of gravity and the elastic axis with respect to the tractor configuration makes this system more susceptible to flutter. The critical velocities for the pusher configuration as a function of the propeller's span position and mounting stiffness are shown in Fig. 13. It is obvious that the flutter speed is drastically reduced for most of the span positions compared to the standalone wing (dashed line), which is due to the mentioned position of the propeller. Reduction in critical speed is observed for a propeller positioned below 70% half-span with a minimized flutter speed at about 30% half-span. The critical speed is higher for propellers positioned toward the outer span positions. This stabilization is so high that for positions near the wing tip, the critical speed is even higher than the standalone wing. This behavior was not expected beforehand, and efforts were made to investigate the origin. It was found that the trend is due to the inertia effect and a favorable change in wind-off eigenfrequencies of the system. In order to check this, an analysis was performed with a nonrotating propeller and without propeller aerodynamics (Fig. 13 - solid black line).

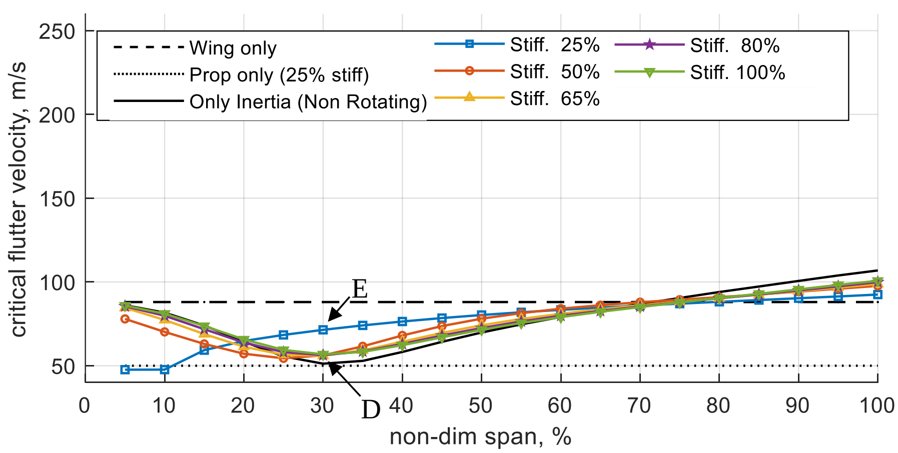


Fig. 13 Critical velocity for a pusher configuration with one attached propeller as a function of span position and mounting stiffness

It is seen that the critical speeds for this system are very similar to the cases with rotation and propeller aerodynamics. Fig. 14 shows the difference between mode 1 and mode 2 eigenfrequencies in wind-off conditions (coupled modes responsible for flutter) for the 100% stiffness case. The proximities of these frequencies are responsible for the trend noted in the flutter behavior. In contrast to the tractor configuration, the mounting stiffness has only a slight influence on the system's stability. For half-span positions higher than 30%, a weaker mounting stiffness tends to have a stabilizing effect. Excluded the smallest mounting stiffness, flutter is due to wing motion. A representative flutter mode shape is shown for a half-span position of 30% and a mounting stiffness of 100% in Fig. 15. As seen, it contains mainly wing motions, and whirl motion is only minor. An exception was found for the very weak mounting case. This mounting stiffness tends to have a stabilizing effect on wing flutter. Instead, whirl flutter is the critical flutter effect. This is illustrated with the flutter curves and flutter mode shape for a half-span position of 30% (Point E) in Fig. 15. In this case, the backward whirl motion is strongly coupled with the wing's pitch. The wing's bending contribution to this mode shape is only minor. Nevertheless, compared to the standalone propeller (Fig. 13 – dashed line), the wing motion stabilizes the whirl flutter, even for the pusher configuration.

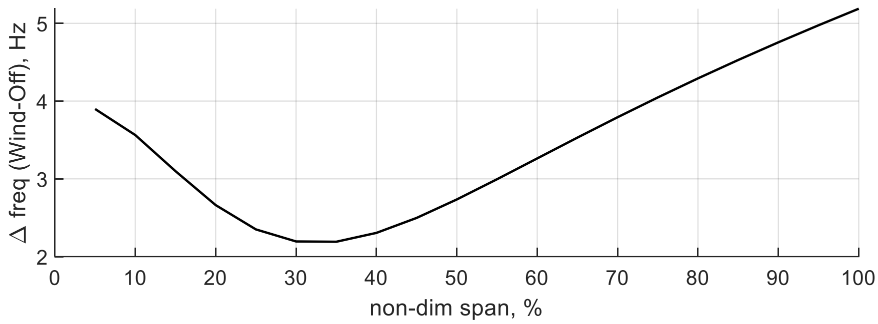


Fig. 14 Difference (abs) in wind-off eigenfrequencies between mode 1 and 2 for 100% mounting stiffness

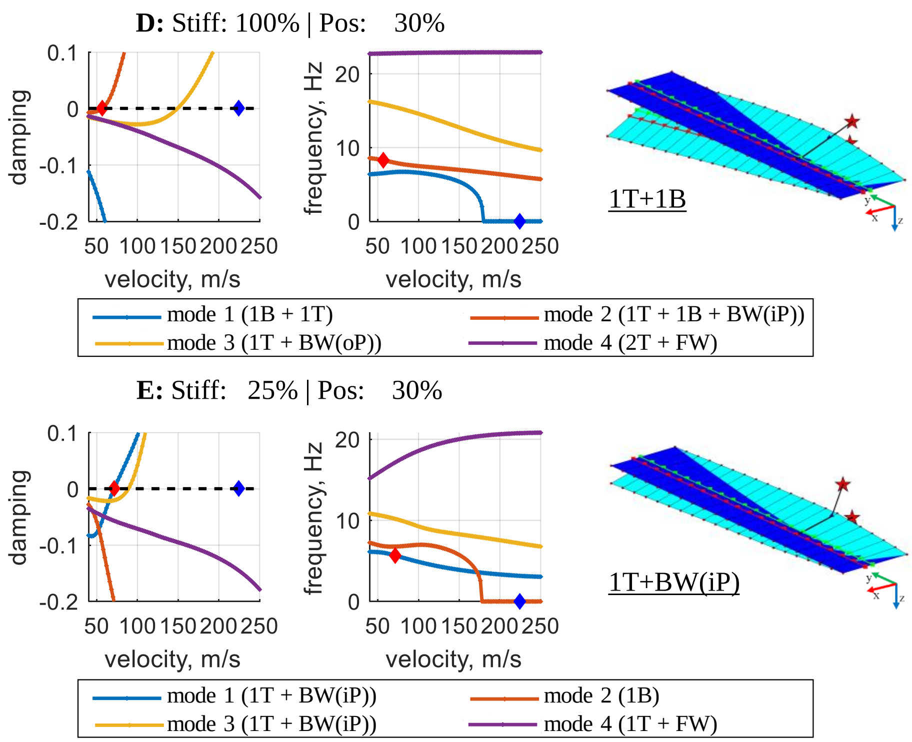


Fig. 15 Flutter curves and flutter mode shape for pusher configuration with the propeller at 30% half-span for two different mounting stiffnesses. Red diamond shows flutter. Blue diamond shows divergence.

VI.Conclusion

Distributed propulsion systems are often used in modern aircraft design. This paper focus on the dynamic aeroelastic stability of two possible wing-propeller configurations: tractor and pusher. The propeller is located in front of the elastic axis in the tractor configuration. In the pusher configuration, the propeller is located behind the trailing edge. Two well-known aeroelastic phenomena can occur in these complex dynamic systems: wing and whirl flutter. Both effects must be avoided during an aircraft operation since they lead to a hazardous failure. According to the certification specifications, the aircraft must be shown to be free from any aeroelastic failure. Especially for rotating systems, such as propellers, it is required to consider possible structural failures that would reduce stiffness and foster whirl flutter. Therefore, the mounting stiffness of the propeller onto the wing is significant and is the scope of studies performed in this paper. Previous studies showed a stabilizing effect with the increasing span position of one propeller attached to the wing. This paper continues this work by changing the span position, mounting stiffness, and chordwise position in the form of a pusher and tractor configuration.

The wing is modeled flexibly and can undergo vibrations in out-of-plane bending and torsion. Unsteady wing aerodynamics are modeled using the doublet-lattice method. Two torsional springs attach the propellers at various span positions to the wing, allowing for pitch and yaw motions. The propeller shaft and blades are assumed to be rigid. The unsteady propeller aerodynamics are modeled using propeller stability derivatives. For the coupled wing-propeller system, aerodynamic interaction is neglected. Due to the propeller's rotation, whirl motions occur and lead to complex coupled mode shapes for the complex wing-propeller model. Aeroelastic instabilities are analyzed using the p-k method in the frequency domain. Critical aeroelastic velocities are extracted and used for comparison of the parameter variation. Efforts were made to assign the appearing flutter mechanism to either wing or whirl flutter by looking into the flutter mode shapes.

A parameter study was performed using the baseline wing featuring one flexible mounted propeller to compare the general system characteristics of a tractor and pusher configuration. The tractor configuration is generally stabilized with respect to the standalone wing, even though this stabilization holds only for higher span positions (i.e., towards the wing tip). A reduction in mounting stiffness will lead to a destabilization and a reduced flutter velocity. The flutter mechanism is mainly due to wing flutter. It was shown that the flutter speed is drastically reduced if the mounting is very weak. In this case, whirl flutter will occur. Nevertheless, corresponding critical velocities are higher than the standalone propeller. In the case of a pusher configuration, the coupled system is much more critical. This was expected since the propeller, with its inertia properties, is attached behind the elastic axis, which fosters aeroelastic effects. The critical speeds are not highly sensitive to the mounting stiffness, except for a very weak mounting. Instead, some span positions were identified as most critical with very low flutter speeds, which is explained by the small difference in the wind-off eigenfrequencies of the coupled modes 1 and 2. Thus, a different wing with changed stiffness distributions would lead to different results. However, it is believed that the general trend holds.

For both configurations, whirl flutter mechanism always occurs with a backward whirl motion that is in phase with the wing pitch motion. In addition, the wing tip is the most stable span position of the propeller for the dynamic stability of the coupled wing-propeller model. However, the pusher configuration is the more critical system due to the inertia effects, which was expected.

It should be mentioned that all results are only valid for the used models and that the focus is on aeroelastic dynamic stability. Of course, a propeller mounted at the wing tip will significantly influence the wing's structural strength since the maximum wing bending moment at the root will be increased. Both aspects, dynamic stability and strength requirements, must be considered for the aircraft design and structural sizing. These aspects will be considered in further studies.

VII. References

[1] W. H. Reed, "Propeller-Rotor Whirl Flutter," Langley Research Center, Symposium on the Noise and Loading Actions on Helicopter V/STOL Aircraft and Ground Effect Machines, Sep. 1965.

[2] W. P. Rodden and T. L. Rose, "Propeller/Nacelle Whirl Flutter Addition to MSC/NASTRAN," Mar. 1989.

[3] Bennett, R. M, Bland, and S. R, "Experimental and analytical investigation of propeller whirl flutter of a power plant on a flexible wing," 1964.

[4] N. Böhnisch, C. Braun, S. Koschel, and P. Marzocca, "A Building-Block Approach to Study Aeroelastic Instabilities for Unconventional Aircraft Configurations," in *19th Australian International Aerospace Congress*, Melbourne, Australia, 2021.

[5] P. C. Teixeira and C. E. Cesnik, "Propeller Influence on the stability of HALE Aircraft," in *31st Congress of the International Council of the Aeronautical Sciences*, Belo Horizonte, Brazil, Sep. 2018.

[6] N. Böhnisch, C. Braun, S. Koschel, M. Vincenzo, and P. Marzocca, "Dynamic Aeroelasticity of Wings with Distributed Propulsion System Featuring a Large Tip Propeller," International Forum on Aeroelasticity and Structural Dynamics, Madrid, Spain, Jun. 2022.

[7] C. Quanglong, H. Jinglong, and Y. Haiwei, "1084. Effect of engine thrust on nonlinear flutter of wings," *Vibroengineering: Journal of Vibroengineering*, vol. 15, no. 4, 2013.

[8] G. P. Guruswamy, "Dynamic Aeroelasticity of Wings with Tip Propeller by Using Navier–Stokes Equations," *AIAA Journal*, vol. 57, no. 8, pp. 3200–3205, 2019, doi: 10.2514/1.J058610.

[9] J. Čečrdle, *Whirl Flutter of Turboprop Aircraft Structures*. United Kingdom: Woodhead Publishing, 2015.

[10] C. Koch, "Parametric whirl flutter study using different modelling approaches," *CEAS Aeronaut J*, vol. 13, no. 1, pp. 57–67, 2022, doi: 10.1007/s13272-021-00548-0.

[11] C. B. Hoover and J. Shen, “Parametric Study of Whirl Flutter Stability of the NASA Aircraft X-57,” in *2018 AIAA Aerospace Sciences Meeting*, Kissimmee, Florida, 2018, doi: 10.2514/6.2018-0276.
[12] N. Böhnisch, C. Braun, S. Koschel, and P. Marzocca, “Whirl Flutter for Distributed Propulsion Systems on a Flexible Wing,” in *AIAA Scitech 2022 Forum*, San Diego, CA & Virtual, 2022, doi: 10.2514/6.2022-1755.
[13] N. Böhnisch and M. Bangel, *DLMpro: Doublet Lattice Method in MATLAB*, 2022. [Online]. Available: https://de.mathworks.com/matlabcentral/fileexchange/110175-dlmpro
[14] A. Savino, A. Cocco, A. Zanotti, M. Tugnoli, P. Masarati, and V. Muscarello, “Coupling Mid-Fidelity Aerodynamics and Multibody Dynamics for the Aeroelastic Analysis of Rotary-Wing Vehicles,” *Energies*, vol. 14, no. 21, p. 6979, 2021, doi: 10.3390/en14216979.
[15] D. Montagnani, M. Tugnoli, F. Fonte, A. Zanotti, M. Syal, and G. Droandi, “Mid-Fidelity Analysis of Unsteady Interactional Aeordynamics of Complex VTOL Configurations,” in *45th European Rotorcraft Forum*, Warsaw, Poland, 2019.
[16] P. Masarati, M. Morandini, and P. Mantegazza, “An Efficient Formulation for General-Purpose Multibody/Multiphysics Analysis,” *Journal of Computational and Nonlinear Dynamics*, vol. 9, no. 4, 2014, doi: 10.1115/1.4025628.
[17] N. Böhnisch, C. Braun, V. Muscarello, and P. Marzocca, “On the Suitability of Conventional Methods for the Aeroelastic Analysis of Novel Aircraft with Distributed Propulsion Systems,” in *20th Australian International Aerospace Congress*, Melbourne, Australia, 2023.
[18] R. Zwaan and H. Bergh, “Propeller-nacelle flutter of the Lockheed Electra aircraft,” Nationaal Luchtvaartlaboratorium Report F.228, Feb. 1962.